



CLASSROOM CONTACT PROGRAMME

(Academic Session : 2021 - 2022)

JEE(Advanced)
REVIEW TEST
18-04-2021

JEE(Main + Advanced) : ENTHUSIAST COURSE (PHASE : TEAS)

PAPER-1

Time : 3 Hours

Maximum Marks : 186

READ THE INSTRUCTIONS CAREFULLY

GENERAL :

1. This sealed booklet is your Question Paper. Do not break the seal till you are told to do so.
2. Use the Optical Response sheet (ORS) provided separately for answering the questions.
3. Blank spaces are provided within this booklet for rough work.
4. Write your name, form number and sign in the space provided on the back cover of this booklet.
5. After breaking the seal of the booklet, verify that the booklet contains **28** pages and that all the **18** questions in each subject and along with the options are legible. If not, contact the invigilator for replacement of the booklet.
6. You are allowed to take away the Question Paper at the end of the examination.

OPTICAL RESPONSE SHEET :

7. The ORS will be collected by the invigilator at the end of the examination.
8. Do not tamper with or mutilate the ORS. **Do not use the ORS for rough work.**
9. Write your name, form number and sign with pen in the space provided for this purpose on the ORS. **Do not write any of these details anywhere else on the ORS.** Darken the appropriate bubble under each digit of your form number.

DARKENING THE BUBBLES ON THE ORS :

10. Use a **BLACK BALL POINT PEN** to darken the bubbles on the ORS.
11. Darken the bubble  **COMPLETELY.**
12. The correct way of darkening a bubble is as : 
13. The ORS is machine-gradable. Ensure that the bubbles are darkened in the correct way.
14. Darken the bubbles **ONLY IF** you are sure of the answer. There is **NO WAY** to erase or "un-darken" a darkened bubble.
15. Take **$g = 10 \text{ m/s}^2$** unless otherwise stated.

Please see the last page of this booklet for rest of the instructions

DO NOT BREAK THE SEALS WITHOUT BEING INSTRUCTED TO DO SO BY THE INVIGILATOR

SOME USEFUL CONSTANTS

Atomic No. : H = 1, B = 5, C = 6, N = 7, O = 8, F = 9, Al = 13, P = 15, S = 16, Cl = 17, Br = 35, Xe = 54, Ce = 58

Atomic masses : H = 1, Li = 7, B = 11, C = 12, N = 14, O = 16, F = 19, Na = 23, Mg = 24, Al = 27, P = 31, S = 32, Cl = 35.5, Ca = 40, Fe = 56, Br = 80, I = 127, Xe = 131, Ba = 137, Ce = 140,

- **Boltzmann constant** $k = 1.38 \times 10^{-23} \text{ JK}^{-1}$
- **Coulomb's law constant** $\frac{1}{4\pi\epsilon_0} = 9 \times 10^9$
- **Universal gravitational constant** $G = 6.67259 \times 10^{-11} \text{ N-m}^2 \text{ kg}^{-2}$
- **Speed of light in vacuum** $c = 3 \times 10^8 \text{ ms}^{-1}$
- **Stefan-Boltzmann constant** $\sigma = 5.67 \times 10^{-8} \text{ Wm}^{-2}\text{-K}^{-4}$
- **Wien's displacement law constant** $b = 2.89 \times 10^{-3} \text{ m-K}$
- **Permeability of vacuum** $\mu_0 = 4\pi \times 10^{-7} \text{ NA}^{-2}$
- **Permittivity of vacuum** $\epsilon_0 = \frac{1}{\mu_0 c^2}$
- **Planck constant** $h = 6.63 \times 10^{-34} \text{ J-s}$

Space for Rough Work

HAVE CONTROL → HAVE PATIENCE → HAVE CONFIDENCE ⇒ 100% SUCCESS

BEWARE OF NEGATIVE MARKING

PART-1 : PHYSICS

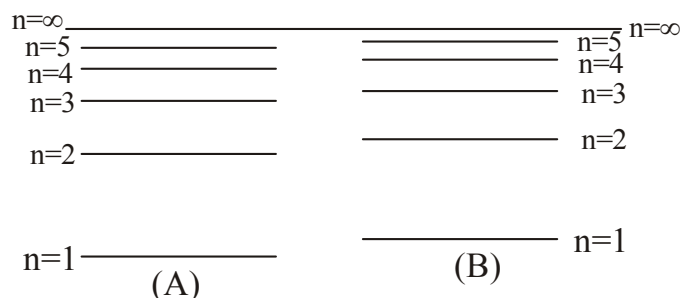
SECTION-I(i) : (Maximum Marks : 12)

- This section contains **FOUR** questions.
- Each question has **FOUR** options (A), (B), (C) and (D). **ONLY ONE** of these four options is correct.
- For each question, darken the bubble corresponding to the correct option in the ORS.
- For each question, marks will be awarded in one of the following categories :
Full Marks : +3 If only the bubble corresponding to the correct option is darkened.
Zero Marks : 0 If none of the bubbles is darkened.
Negative Marks : -1 In all other cases

-
1. Suppose that at the equator of the Earth we built a tower whose height $H = 2.5 \times 10^4$ km. If an object was dropped from the top of the tower,
- (A) It would fall to the earth in a straight line
 - (B) It would orbit the earth in a circular path
 - (C) It would orbit the earth in an elliptical path with the initial position as the closest position from the centre of the earth
 - (D) It would orbit the earth in an elliptical path with the initial position as the farthest position from the centre of the earth
-

Space for Rough Work

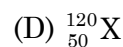
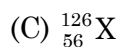
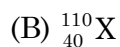
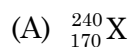
2. The quantum levels of atoms of hydrogen and deuterium are only approximately the same (the difference between the two systems of levels is exaggerated in the figure).



Statement-1 : A corresponds to quantum levels of hydrogen atom and B to quantum levels of deuterium atom.

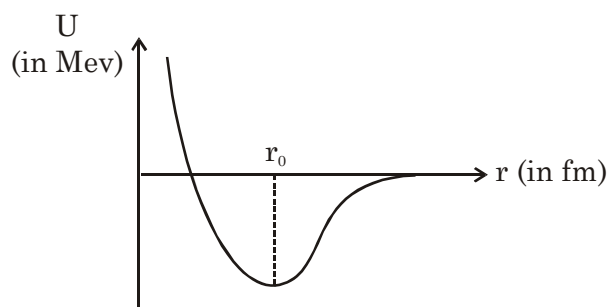
Statement-2 : The mass of H nuclei is lesser than the mass of deuterium nuclei.

- (A) Statement-1 is true, statement-2 is true and statement-2 is correct explanation for statement-1.
 (B) Statement-1 is true, statement-2 is true and statement-2 is NOT the **CORRECT** explanation for statement-1.
 (C) Statement-1 is true, statement-2 is false.
 (D) Statement-1 is false, statement-2 is true.
3. An unknown nucleus contains 70 neutrons and has twice the volume of the ${}^{60}_{28}\text{Ni}$ nickel nucleus. Identify the unknown nucleus in the form .



Space for Rough Work

4. The potential energy of interaction between 2 nucleons as a function of distance is shown here in the graph. The distance r_0 at which the potential energy is minimum is approximately:



(A) 0.8 fm

(B) 0.9 fm

(C) 1.0 fm

(D) 1.1 fm

Space for Rough Work

SECTION-I(ii) : (Maximum Marks: 32)

- This section contains **EIGHT** questions.
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- For each question, choose the correct option(s) to answer the question.
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- **For Example :** If first, third and fourth are the **ONLY** three correct options for a question with second option being an incorrect option; selecting only all the three correct options will result in +4 marks. Selecting only two of the three correct options (e.g. the first and fourth options), without selecting any incorrect option (second option in this case), will result in +2 marks. Selecting only one of the three correct options (either first or third or fourth option), without selecting any incorrect option (second option in this case), will result in +1 marks. Selecting any incorrect option(s) (second option in this case), with or without selection of any correct option(s) will result in -1 marks.

5. Consider a particle confined in a one-dimensional box of length L . The potential energy in the region outside the box is infinite. The box can be divided into two parts. The potential energy in the left half of the box is U_0 , and the potential energy of the particle in the right half of the box is zero. The mass of the particle is m . What can be the possible values of U_0 so that standing wave pattern of matter wave is formed inside the box with node at mid-point of the box ?

(A) $\frac{h^2}{2mL^2}$ (B) $\frac{h^2}{8mL^2}$ (C) $\frac{7h^2}{2mL^2}$ (D) $\frac{6h^2}{mL^2}$

Space for Rough Work

-
- Diagram illustrating a circle with center O . A horizontal chord connects points A (labeled 'New Delhi') and B (labeled 'Moscow'). The central angle $\angle AOB$ is marked with a dashed arc and labeled $\pi R/3$. The minor arc AB is marked with a solid arc and labeled $\pi R/3$.

- (A) The initial acceleration of the train as it enters the tunnel is 5 m/s^2
 (B) The time taken from New Delhi to Moscow is approx. 45 min
 (C) The speed of the rail is maximum at the centre of the tunnel
 (D) The normal force on the rails is maximum at the initial position.
7. In an X ray tube, the applied potential difference is V . If the emitted x rays are incident on a hydrogen like atom of atomic number Z , it can be excited to a higher excited state but cannot be ionized. Select the proper combination of V and Z
- (A) $Z = 11, V = 1250 \text{ V}$ (B) $Z = 9, V = 800 \text{ V}$
 (C) $Z = 10, V = 1400 \text{ V}$ (D) $Z = 5, V = 320 \text{ V}$

Space for Rough Work

8. Just after a K shell capture, which of the following come from the atom ?
 (A) Neutrino (B) Antineutrino (C) x-Ray photon (D) Electron

9. In the reaction ${}^2_1\text{H} + {}^2_1\text{H} \longrightarrow {}^3_1\text{H} + {}^1_1\text{H}$

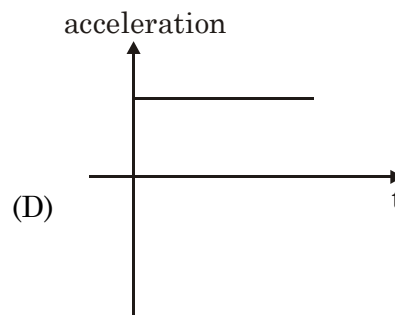
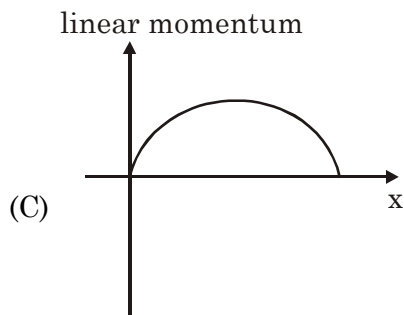
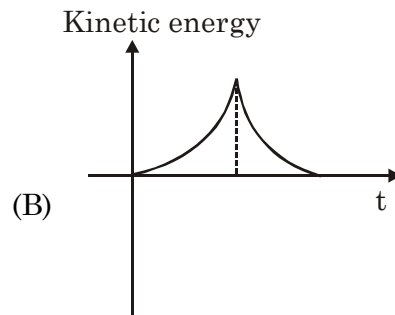
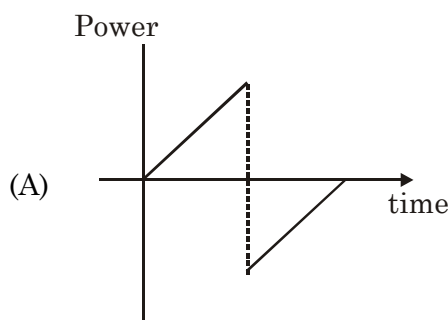
The Q value is 4.0 MeV, and so this nuclear reaction liberates about 1 MeV per nucleon, roughly the same as the fission reaction. This reaction can be performed in the laboratory, by accelerating a beam of deuterons on to a deuterium target. mass of hydrogen atom = 1.0078 amu. Mass of Deuterium atom = 2.0141 amu, 1 amu = 931.5 MeV/c², Radius of deuterium nucleus is 1.5 fm.

Select the **CORRECT** option(s)

- (A) For initiating the reaction, the energy of incident deuterons should be of the order of 1.0 MeV
 (B) For initiating the reaction, the energy of incident deuterons should be of the order of 8.0 MeV
 (C) The mass of tritium atom is 3.0161 amu
 (D) The mass of tritium atom is 3.0204 amu
10. For a certain radioactive substance, it is observed that after 4 hours, only 6.25% of the original sample is left undecayed. It follows that
 (A) the half life of the sample is 1 hour
 (B) the mean life of the sample is $\frac{1}{\ln 2}$ hour
 (C) the decay constant of the sample is $\ln 2 \text{ hour}^{-1}$
 (D) after a further 4 hours, the amount of the substance left over would be only 0.39% of the original amount

Space for Rough Work

11. A block is moving with constant velocity on the fixed rough inclined plane having angle of inclination θ then choose the **CORRECT** option :
- (A) In a fixed time interval work done by weight and friction are equal on the block
- (B) $\tan\theta = \mu$
- (C) Work done by all forces on the block is zero.
- (D) Mechanical energy of block-earth system is decreasing.
12. An object initially at rest at origin is acted upon by a constant force along x-axis. After some time the direction of force is reversed and object is finally brought to rest again then choose the **CORRECT** option :-



Space for Rough Work

SECTION-II : (Maximum Marks: 18)

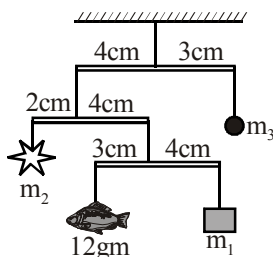
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- The answer to each question is a **NUMERICAL VALUE**.
- For each question, enter the correct numerical value (in decimal notation, truncated/rounded-off to the **second decimal place**; e.g. 6.25, 7.00, -0.33 , -0.30 , 30.27, -127.30 , if answer is 11.36777..... then both 11.36 and 11.37 will be correct) by darkening the corresponding bubbles in the ORS.
For Example : If answer is -77.25 , 5.2 then fill the bubbles as follows.

		+		-	
0	0	0	0	0	0
1	1	1	1	1	1
2	2	2	2	2	2
3	3	3	3	3	3
4	4	4	4	4	4
5	5	5	5	5	5
6	6	6	6	6	6
7	7	7	7	7	7
8	8	8	8	8	8
9	9	9	9	9	9

		+		-	
0	0	0	0	0	0
1	1	1	1	1	1
2	2	2	2	2	2
3	3	3	3	3	3
4	4	4	4	4	4
5	5	5	5	5	5
6	6	6	6	6	6
7	7	7	7	7	7
8	8	8	8	8	8
9	9	9	9	9	9

- Answer to each question will be evaluated according to the following marking scheme:
Full Marks : +3 If **ONLY** the correct numerical value is entered as answer.
Zero Marks : 0 In all other cases.

1. A monochromatic source of light of wavelength 200 nm is kept at the centre of curvature of a perfectly reflecting mirror of radius of curvature 20 cm and aperture diameter also 20 cm. The power of the source is 4800 W. What is the force (in μN) experienced by the mirror due to radiation pressure ?
2. A souvenir is constructed by 3 light rods, light strings and four objects as shown. If the system is in equilibrium, what is mass m_3 (in gm)

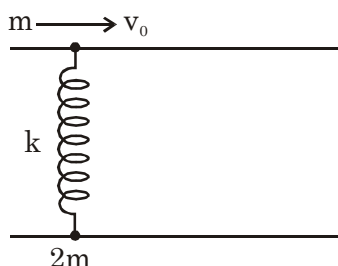


Space for Rough Work

3. An electric dipole (dipole moment = p) is placed in a uniform electric field in stable equilibrium position at rest. Now it is rotated by a small angle and released. The time after which it comes to the equilibrium position again (for first time) is t . If the moment of inertia of the dipole about the axis of rotation is $x \frac{t^2 p E}{\pi^2}$, then find the value of x .
4. Flat aluminum electrode is illuminated by light with a wavelength of 83 nm. What is the maximum distance (in mm) from the electrode surface will a photoelectron move, if outside the electrode there is a retarding electric field of $E = 750$ V/m. Photoelectric threshold for aluminum is 332 nm. Take $hc = 1245$ eV nm
-

Space for Rough Work

5. Two ring of mass m and $2m$ are connected with a light spring and can slide over two frictionless parallel horizontal rails as shown in figure. Ring of mass m is given velocity ' v_0 ' in horizontal direction as shown. If the maximum stretch in spring during subsequent motion is $\sqrt{n \frac{mv_0}{k}}$, then find the value of n .



6. A fixed circle of radius a and centre O is drawn and a charge $+q$ is placed at a distance $3a/4$ from O on a line through O perpendicular to the plane of circle. If a second charge q' is similarly placed at a distance $\frac{5a}{12}$ on the opposite side of circle, the net electric flux through circle becomes zero. Find the ratio of q' to q .

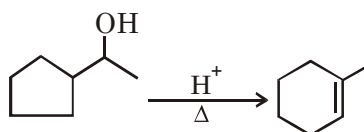
Space for Rough Work

PART-2 : CHEMISTRY

SECTION-I(i) : (Maximum Marks : 12)

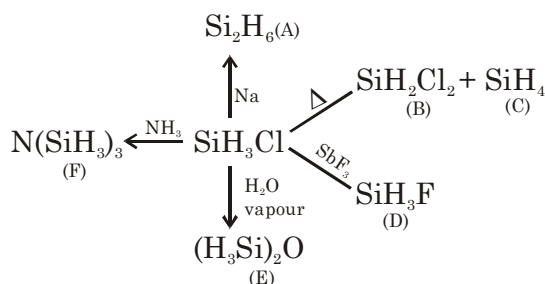
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1. In the following reaction :



Find out number of 1,2-shift of the carbocation formed :

- (A) 2 (B) 1 (C) 3 (D) 4
2. Consider the following reactions of SiH_3Cl .



Which of the following statement is **INCORRECT**?

- (A) Atleast one bond angle of (B) is greater than every bond angle of (C) and (D).
 (B) E is more basic than F
 (C) All Si compounds in the above diagram have Si in sp^3 hybridization
 (D) Atleast one out of (A), (B), (C), (D), (E), (F) shows back bonding & bridge bonding simultaneously

Space for Rough Work

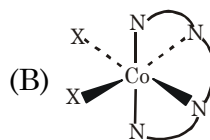
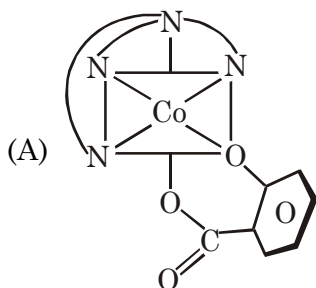
3. Consider the equilibrium $\text{Ni(s)} + 4\text{CO(g)} \rightleftharpoons \text{Ni(CO)}_4\text{(g)}$; $K_p = \frac{1}{16} \text{ atm}^{-3}$. If ideal gases CO and Ni(CO)_4 are mixed in a small container fitted with a piston in 1:4 mole ratio then calculate the maximum total pressure to which this mixture must be brought in order to just precipitate out metallic Ni.
- (A) 760 cm Hg (B) 1520 cm Hg
(C) 2280 cm Hg (D) 3040 cm Hg
4. The relative nucleophilicity order in polar protic solvents (H_2O) of the following nucleophile is :
- (a) $\text{CH}_3\text{CH}_2\ddot{\text{S}}^-$ (b) $\text{CH}_3\text{CH}_2\text{O}^-$ (c) CH_3CO_2^-
- (A) $a > b > c$ (B) $b > c > a$ (C) $a > c > b$ (D) $c > a > b$

Space for Rough Work

SECTION-I(ii) : (Maximum Marks: 32)

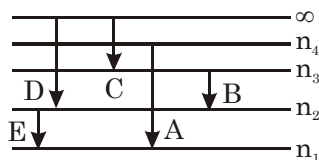
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5. Which of the following complex(es) can show geometrical isomerism.



Space for Rough Work

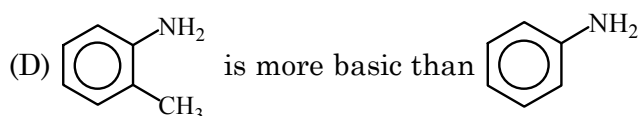
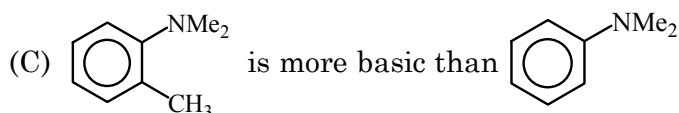
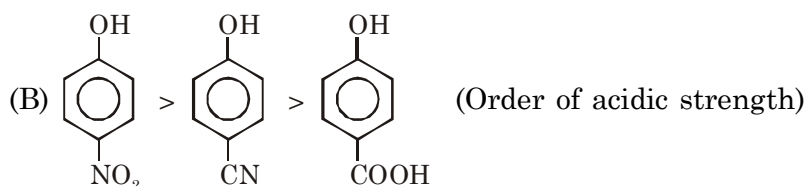
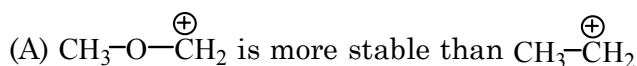
6. For a hypothetical H like atom which follows Bohr model, some spectral lines were observed as shown in figure. If it is known that line E belongs to visible region, then the lines possibly belonging to ultraviolet region will be : [Given : n_1, n_2, n_3, n_4 are consecutive orbits and $n_1 = 3$]



- (A) line B (B) line C
(C) line D (D) line A
7. Correct statement regarding following compound is :
-
- (A) Double bond equivalent (D.B.E.) is 6
(B) Compound mainly exist in its enol form
(C) Enol form is more acidic than keto form
(D) Enol form contain lower number of p-orbital in resonance than keto form
8. Specie(s) that have square planar geometry around underlined atom is/are.
(A) $\underline{\text{I}}_2\text{Cl}_6$ (B) $[\underline{\text{Pd}}\text{Cl}_4]^{2-}$ (C) $[\underline{\text{Pd}}_2\text{Cl}_4(\text{PPh}_3)_2]$ (D) $[\underline{\text{Fe}}\text{O}_4]^{2-}$
9. In a hypothetical 1st order decomposition reaction $\text{A(aq)} \rightleftharpoons 2\text{B(aq)} + \text{C(aq)}$, 'A' is optically active (dextrorotatory) while 'B' and 'C' are optically inactive but 'B' takes part in a titration reaction with H_2O_2 . Hence the progress of reaction can be monitored by measuring rotation of plane polarized light or by measuring volume of H_2O_2 consumed in titration. In an experiment the optical rotation was found to be $\theta = 40^\circ$ at $t = 20$ min and $\theta = 10^\circ$ at $t = 50$ min from start of the reaction. If the progress would have been monitored by titration method, volume of H_2O_2 consumed at $t = 15$ min (from start) is 40 ml then select the correct option :
- (A) Volume of H_2O_2 consumed at $t = 60$ min will be 70 ml.
(B) Volume of H_2O_2 consumed at $t = 60$ min will be 75 ml.
(C) Optical rotation was found to be $\theta = 20^\circ$ at $t = 40$ min
(D) Optical rotation was found to be $\theta = 20^\circ$ at $t = 35$ min

Space for Rough Work

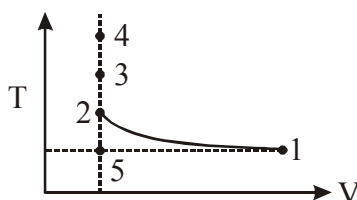
10. Select correct statement among following -



11. Which of the following statement(s) is/are **CORRECT**?

- (A) Conduction in metals is due to movement of electrons from valence band (filled molecular orbitals) to conduction band (unfilled molecular orbitals)
- (B) Only π^* molecular orbitals are responsible for synergic bonding in complexes having non-classical ligands
- (C) Due to low separation in energies of HOMO-LUMO in F_2 compared to I_2 , the intensity of color is lower in F_2
- (D) In ionization of $\text{CO} \rightarrow \text{CO}^+$, the bond length of C-O bond decreases

12. If the plot $1 \rightarrow 2$ represents an infinite stage adiabatic process and $1 \rightarrow 3$ represents a single stage adiabatic process, then identify the incorrect statement



- (A) An intermediate state between point 3 and 4 can be reached by an adiabatic process.
- (B) An intermediate state between point 2 and 5 can be reached by an adiabatic process.
- (C) An intermediate state between point 3 and 2 can be reached by an adiabatic process.
- (D) Points 4 and 5 can lie on any adiabatic process.

Space for Rough Work

SECTION-II : (Maximum Marks: 18)

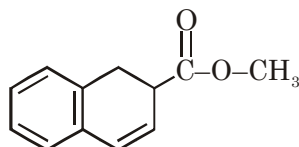
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3	3	3	3	3	3	3	3	3	3
4	4	4	4	4	4	4	4	4	4
5	5	5	5	5	5	5	5	5	5
6	6	6	6	6	6	6	6	6	6
7	7	7	7	7	7	7	7	7	7
8	8	8	8	8	8	8	8	8	8
9	9	9	9	9	9	9	9	9	9

- Answer to each question will be evaluated according to the following marking scheme:
Full Marks : +3 If **ONLY** the correct numerical value is entered as answer.
Zero Marks : 0 In all other cases.

1. How many sp^2 carbons are present in following compound :



2. Among the species



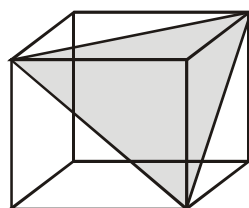
if $X \rightarrow$ number of paramagnetic species

$Y \rightarrow$ number of species having pi bond

then number of degenerate orbitals in $\left(\frac{X+Y}{2}\right)^{\text{th}}$ excited state of Li^{2+} is _____.

Space for Rough Work

3. The volume of the shaded space of a cube of edge length 3cm in cm^3 is:



4. Total number of possible alkadienes structures having the molecular formula C_5H_8 is (Excluding stereoisomers) :
5. Specie(s) that share atleast one atom of one tetrahedral unit with other tetrahedral unit among the following.
 $\text{Si}_2\text{O}_7^{2-}$, $\text{Si}_4\text{O}_{11}^{6-}$, B_2H_6 , Al_2Cl_6 , I_2Cl_6 , $\text{Na}_6\text{P}_6\text{O}_{18}$, N_2H_4
6. An ideal gas undergoes a process in which $T = 300 + aV^3$. Where V is the molar volume in Liter/mole, T is absolute temperature in Kelvin and a is a constant. Find the molar volume in liter/mole at which pressure will be minimum if $a = 1.2 \text{ Kmol}^3\text{L}^{-3}$.

Space for Rough Work

PART-3 : MATHEMATICS**SECTION-I(i) : (Maximum Marks : 12)**

- This section contains **FOUR** questions.
- Each question has **FOUR** options (A), (B), (C) and (D). **ONLY ONE** of these four options is correct.
- For each question, darken the bubble corresponding to the correct option in the ORS.
- For each question, marks will be awarded in one of the following categories :
Full Marks : +3 If only the bubble corresponding to the correct option is darkened.
Zero Marks : 0 If none of the bubbles is darkened.
Negative Marks : -1 In all other cases

1. $\lim_{x \rightarrow -\infty} \left(\sqrt{x^2 - \frac{4x}{3}} + x + 2 \right)$ is equal to-
 (A) $\frac{4}{3}$ (B) $-\frac{4}{3}$ (C) $\frac{8}{3}$ (D) $-\frac{8}{3}$
2. If a derivable function $y = f(x)$ satisfies the condition $f(x) + f(y) = f(x)f(y) + f(xy) \forall x, y \in \mathbb{R}$ where $f(1) = 0$ and $f'(1) = -2$, then the number of solution(s) of the equation $f(x) = 2^x$ is
 (A) 4 (B) 3 (C) 2 (D) 1
3. The number of integral values of k for which the equation $e^x = \frac{k}{x-3}$, has exactly two solutions is-
 (A) 5 (B) 6 (C) 7 (D) 9
4. Let $F(x) = C \sin x \int_0^x \cos t dt + 2 \int_0^x t dt + C \cos^2 x - x^2$ (where $C \in \mathbb{R}^+$). If $x^2 - 2x + 3 \geq F(x) \forall x \in \mathbb{R}$, then the maximum value of the function $\left(\frac{6e^x - e^{2x} - 1}{e^x} \right)^{F(2)}$ is equal to -
 (A) 2 (B) 4 (C) 16 (D) 32

Space for Rough Work

SECTION-I(ii) : (Maximum Marks: 32)

- This section contains **EIGHT** questions.
- Each question has **FOUR** options for correct answer(s). **ONE OR MORE THAN ONE** of these four option(s) is (are) correct option(s).
- For each question, choose the correct option(s) to answer the question.
- Answer to each question will be evaluated according to the following marking scheme:
Full Marks : +4 If only (all) the correct option(s) is (are) chosen.
Partial Marks : +3 If all the four options are correct but **ONLY** three options are chosen.
Partial Marks : +2 If three or more options are correct but **ONLY** two options are chosen, both of which are correct options.
Partial Marks : +1 If two or more options are correct but **ONLY** one option is chosen and it is a correct option.
Zero Marks : 0 If none of the options is chosen (i.e. the question is unanswered).
Negative Marks : -1 In all other cases.
- **For Example** : If first, third and fourth are the **ONLY** three correct options for a question with second option being an incorrect option; selecting only all the three correct options will result in +4 marks. Selecting only two of the three correct options (e.g. the first and fourth options), without selecting any incorrect option (second option in this case), will result in +2 marks. Selecting only one of the three correct options (either first or third or fourth option), without selecting any incorrect option (second option in this case), will result in +1 marks. Selecting any incorrect option(s) (second option in this case), with or without selection of any correct option(s) will result in -1 marks.

5. If $\int \frac{x + \sqrt[3]{x^2} + \sqrt[6]{x}}{x(1 + \sqrt[3]{x})} dx = A.x^B + C \tan^{-1}(x^D) + E$, then which of the following is/are true (where 'E' is constant of integration) ?

(A) $A = \frac{2}{3}$

(B) $AB = 1$

(C) $(AB + C)$ is a prime number

(D) Sum of all coefficients of even power of x in the expansion of $\left(AB + \frac{x}{D}\right)^C$ is equal to

$$\frac{(C+1)^C + (C-1)^C}{2}$$

Space for Rough Work

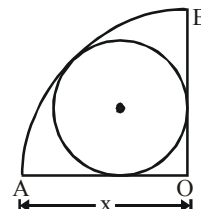
6. The function $f(x) = [\cos x]\cos x$ in $(0, 2\pi)$ -
 (A) is discontinuous at two points (B) is non differentiable at two points
 (C) has range $(0, 1]$ (D) is such that $f(f(x)) + 1 = 0$ has no real roots.
7. Sector OAB is a quarter of circle of radius x . A circle of radius ' r ' is drawn inside this sector tangent to three points as shown, then which of the following is/are true ? (where r & x are variable)

(A) $\left(\frac{dr}{dx}\right)^2 + \left(\frac{dx}{dr}\right)^2 + 2\frac{d^2x}{dr^2} = 6$

(B) $\left(\frac{dr}{dx}\right)^2 + \left(\frac{dx}{dr}\right)^2 - 2\frac{d^2x}{dr^2} = 8$

(C) $\frac{x+r}{x-r} + \frac{r-x}{x+r} = 2$

(D) $\sum_{n=1}^{10} \frac{d^n r}{dx^n} = \frac{r}{x}$



8. Let $f(x) = \begin{cases} -2 & \text{for } -3 \leq x \leq 0 \\ x-2 & \text{for } 0 < x \leq 3 \end{cases}$, $g(x) = [f(|x|)] + [f(x)]$ where $[.]$ denotes the greatest integer function. Which of the following statement(s) is/are true ?
 (A) $g(1.7) = -1$
 (B) $g(x)$ is not differentiable at any point
 (C) $g(x)$ is continuous at all non integral points
 (D) $g(x)$ is differentiable at some points

Space for Rough Work

9. Consider $f(x) = (x-1)\tan^{-1}x - \frac{1}{2}\ln(1+x^2)$, $x \geq 1$. Identify which of the following statement(s) is(are) correct ?
- (A) $\lim_{x \rightarrow \infty} f'(x) = \frac{\pi}{2}$
- (B) $f'(x)$ is increasing in $[1, \infty)$
- (C) $f(x+4) - f(x) > 2\pi \forall x \in [1, \infty)$
- (D) $f(x+4) - f(x) < 2\pi \forall x \in [1, \infty)$
10. Locus of the middle points of chords of parabola $y^2 = 4x$, tangents drawn at the ends of which intersect at right angle is a parabola whose
- (A) Vertex is $(1, 0)$
- (B) Length of the shortest focal chord is 2
- (C) Equation of directrix is $x = \frac{1}{2}$
- (D) tangents at ends of latus rectum intersect at $\left(\frac{1}{2}, 0\right)$

Space for Rough Work

11. If the solution of the equation, $\sin \tan^{-1} x = \sqrt{4 - \sin \cos^{-1} x \cos \sin^{-1} x}^2$ is 'a', then

(A) $\sin^{-1} a \cos^{-1} a = \frac{\pi}{2}$

(B) $2\sin^{-1} a \cos^{-1} a = \frac{\pi}{2}$

(C) $\sin^{-1} a + 3\cos^{-1} a = \frac{3\pi}{2}$

(D) $\tan^{-1} a \cos^{-1} a = \frac{\pi}{2}$

12. If a function $f : R \rightarrow R$ satisfies $\left| \sum_{k=1}^n 3^k [f(x+ky) - f(x-ky)] \right| \leq 1$ for every $n \in N$ and for all $x, y \in R$,

then which of the following is/are true

(A) $\left| 3^n [f(x+ny) - f(x-ny)] \right| \leq 2$

(B) $|f(u) - f(v)| \leq \frac{2}{3^n}$

(C) $f(x)$ is a constant function

(D) $f'(x) = 1 \forall x \in R$

Space for Rough Work

SECTION-II : (Maximum Marks: 18)

- This section contains **SIX** questions.
- The answer to each question is a **NUMERICAL VALUE**.
- For each question, enter the correct numerical value (in decimal notation, truncated/rounded-off to the **second decimal place**; e.g. 6.25, 7.00, -0.33, -.30, 30.27, -127.30, if answer is 11.36777..... then both 11.36 and 11.37 will be correct) by darkening the corresponding bubbles in the ORS.

For Example : If answer is -77.25, 5.2 then fill the bubbles as follows.

+									
●	●	○	○	○	○	○	○	○	○
①	①	①	①	①	①	①	①	①	①
②	②	②	②	●	②	②	②	②	②
③	③	③	③	③	③	③	③	③	③
④	④	④	④	④	④	④	④	④	④
⑤	⑤	⑤	⑤	⑤	⑤	⑤	⑤	⑤	⑤
⑥	⑥	⑥	⑥	⑥	⑥	⑥	⑥	⑥	⑥
⑦	⑦	●	●	⑦	⑦	⑦	⑦	⑦	⑦
⑧	⑧	⑧	⑧	⑧	⑧	⑧	⑧	⑧	⑧
⑨	⑨	⑨	⑨	⑨	⑨	⑨	⑨	⑨	⑨

-									
●	●	●	○	○	○	○	○	○	○
①	①	①	①	①	①	①	①	①	①
②	②	②	②	●	②	②	②	②	②
③	③	③	③	③	③	③	③	③	③
④	④	④	④	④	④	④	④	④	④
⑤	⑤	⑤	●	⑤	⑤	⑤	⑤	⑤	⑤
⑥	⑥	⑥	⑥	⑥	⑥	⑥	⑥	⑥	⑥
⑦	⑦	⑦	⑦	⑦	⑦	⑦	⑦	⑦	⑦
⑧	⑧	⑧	⑧	⑧	⑧	⑧	⑧	⑧	⑧
⑨	⑨	⑨	⑨	⑨	⑨	⑨	⑨	⑨	⑨

- Answer to each question will be evaluated according to the following marking scheme:
Full Marks : +3 If **ONLY** the correct numerical value is entered as answer.
Zero Marks : 0 In all other cases.

1. If θ is the acute angle between the curves $y = |x^2 - 1|$ and $y = |x^2 - 3|$ at their point of intersection when $x > 0$, then $7 \tan \theta$ is equal to
2. Let $f(x)$ be a twice differentiable function satisfying $xf(x^2) + (x^2 + 1)f(x) = x$, $\forall x \in \mathbb{R}$ then the value of $|f''(1)|$ is
3. If $\int_0^1 \frac{e^x dx}{\sqrt{1-x^2}} = A$ and $\int_0^\pi (e^{|\sin x|} + e^{|\cos x|}) dx$ is equal to λA , then the value of λ is -

Space for Rough Work

4. If $f(x) = \begin{cases} \frac{\sin^{-1}(\{x\}^2 + \{x\}^3) - x}{\{x\}}, & x \neq 0 \\ -1, & x = 0 \end{cases}$, then the absolute value of RHD of $f(x)$ at $x = 0$, is

(where $\{.\}$ denotes fractional part function)

5. If the line $x + y + 1 = 0$ and $y - 2x + 5 = 0$ are tangents to a parabola whose focus at $(1, 2)$ then the equation of normal to the parabola through $\left(\frac{29}{17}, \frac{-14}{17}\right)$ is $ax + y = c$, then $(a + c)$ is

6. Let $f(x) = ax^9 + b \sin x + cx^2 \operatorname{sgn} x + \frac{e^x - e^{-x}}{e^x + e^{-x}}$ be defined on set of real number (where $a > 0, b, c \in \mathbb{R}$).

If $f(-5) = 5, f(-2) = -3$ and K be minimum number of zeroes of the equation $f(x) = 0$, then $\frac{K}{4}$ is :

Space for Rough Work

Space for Rough Work

QUESTION PAPER FORMAT AND MARKING SCHEME :

16. The question paper has three parts : Physics, Chemistry and Mathematics.

17. Each part has two sections as detailed in the following table.

Section	Que. Type	No. of Que.	Category-wise Marks for Each Question				Maximum Marks of the section
			Full Marks	Partial Marks	Zero Marks	Negative Marks	
I(i)	Single correct option	4	+3 If only the bubble corresponding to the correct option is darkened	—	0 If none of the bubbles is darkened	–1 In all other cases	12
I(ii)	One or more correct option(s)	8	+4 If only the bubble(s) corresponding to all the correct option(s) is(are) darkened	+1 For darkening a bubble corresponding to each correct option, provided NO incorrect option darkened	0 If none of the bubbles is darkened	–1 In all other cases	32
II	Numerical Value Type (Up to second decimal place)	6	+3 If only the bubble corresponding to correct answer is darkened	—	0 In all other cases	—	18

NAME OF THE CANDIDATE

FORM NO.

I have read all the instructions and shall abide by them.

Signature of the Candidate

I have verified the identity, name and Form number of the candidate, and that question paper and ORS codes are the same.

Signature of the Invigilator